SPAR: Semantic-Pixel Self-Alignment and Adaptive Routing for Unified Multimodal Models

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Abstract:
Multimodal Large Language Models (MLLMs) have achieved remarkable success in visual understanding but remain constrained in visual generation due to the fundamental feature discrepancy between semantic perception and pixel-level reconstruction. Bridging this gap requires overcoming two core challenges: endowing semantic encoders with high-fidelity reconstruction capabilities, and effectively aligning generative models with semantic spaces without relying on external teachers. To this end, we propose a novel unified multimodal framework featuring \textbf{S}emantic-\textbf{P}ixel self-alignment and \textbf{A}daptive \textbf{R}outing (\textbf{SPAR}). First, to reconcile semantic perception with pixel-level reconstruction, we introduce an asymmetric dual-stream unified tokenizer. A lightweight semantic stream anchors discriminative features, while a Transformer-augmented pixel stream recovers fine-grained visual details into a unified compact latent space. Second, to eliminate external dependencies, we propose a self-aligned generation paradigm that natively leverages this optimized tokenizer as an internal alignment teacher for the diffusion model. Furthermore, to facilitate flexible multimodal interaction within this unified space, we introduce Dynamic Token Routing, which enables each token to adaptively aggregate multi-layer MLLM features based on its distinct semantic demands. Extensive experiments demonstrate that SPAR establishes the state-of-the-art for unified architectures, achieving exceptional generation and reconstruction quality while preserving foundational visual understanding capabilities.

Supplementary Material: pdf (/attachment?id=JecfyVt1bP&name=supplementary_material)
Submission Number: 5523

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Review for SPAR

Official Review by Reviewer XtYZ 22 Apr 2026, 03:09 (modified: 03 May 2026, 06:52)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Authors

Revisions (/revisions?id=WixXC2tmWM)

Choose The Contribution Type You Are Using For Your Review: Applied/Systems

Justify The Above Choice:

The paper is more of a training recipe for solving the degradation of semantic encoders for generation. Hence I believe that the paper falls under the category of Applied / Systems.

Paper Summary:

The paper tackles a known pain point in unified MLLMs: semantic encoders are great for understanding but lose pixel detail needed for generation. The authors propose SPAR, which has three main pieces: (1) an asymmetric dual-stream tokenizer where a lightweight semantic branch preserves the frozen encoder's features while a heavier Transformer-based pixel branch recovers high-frequency details, (2) a self-alignment paradigm where the tokenizer itself acts as the alignment teacher for the DiT (replacing external encoders like DINOv2), and (3) Dynamic Token Routing, which lets each token pick its own weighted mix of MLLM layers rather than using only the last layer. Experiments on ImageNet reconstruction, GenEval, WISE, ImgEdit, and several understanding benchmarks show SOTA or near-SOTA results among unified models.

Paper Strengths:

- The asymmetric design is well-motivated. Putting the heavy lifting on the pixel stream while keeping the semantic stream as a light anchor is a clean way to frame the perception-vs-reconstruction tradeoff, and the ablations (Rows b and c in Table 5) back this up convincingly — dropping either stream breaks one side badly.
- Using the tokenizer as its own alignment teacher is an elegant idea. It removes the awkward dependency on an external DINOv2-style model and keeps everything in one representation space.
- Results are strong and broad. The model does well on reconstruction (rFID 0.27), understanding (matches or beats InternVL3), generation (GenEval 0.91), and editing (ImgEdit 4.01, close to GPT-4o) — and does so at a relatively modest parameter count.
- DTR is a reasonable contribution with a nice interpretability angle (per-token layer preferences), though it's more of an incremental addition than the central idea.
- The three-stage training recipe for the tokenizer is clearly laid out and the semantic-distillation loss in Stage III directly addresses catastrophic forgetting.

Major Weaknesses:

- The novelty of individual components is moderate. Dual-stream tokenizers, representation alignment losses (REPA-style), and layer-wise feature aggregation all exist in prior work. The contribution is really the combination and the self-alignment twist, which should be acknowledged more clearly.
- DTR ablation is thin. Table 6 compares against only "last-layer" extraction. A fairer comparison would include static multi-layer concatenation or learned-but-token-agnostic weights, which are the actual baselines DTR claims to beat.
- No discussion of compute or training cost. Adding a 6-layer Transformer in the pixel stream plus a three-stage tokenizer training plus a three-stage unified training is non-trivial, and there's no efficiency comparison with simpler baselines.

Minor Weaknesses:

Some presentation issues. Figure 1(b) is a bit under-explained, the text has a typo ("self-lignment" in the conclusion), and reference [3] pointing to a blog post as evidence that external alignment is suboptimal is a weak citation for a fairly load-bearing claim. Image editing results are impressive but the benchmark (ImgEdit) alone isn't enough — GEdit-Bench or similar would strengthen the claim.

Preliminary Recommendation: 3: Borderline Reject
Justification Of Preliminary Recommendation:
I believe that the ideas are more integrative than fundamentally new - therefore I am going with Borderline Reject. I am happy to revisit my opinion after the rebuttal phase.

Suggestions For Rebuttal:
Please see Weakness.

Ethics Review Flag: No
Confidence Level: 3: Moderate Confidence - The reviewer is reasonably knowledgeable about the topic. They understand the paper's methodology and results but may not be a leading expert in the specific subfield.

Review

Official Review by Reviewer F1M7 21 Apr 2026, 23:59 (modified: 03 May 2026, 06:52)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Authors
Revisions (/revisions?id=yN36yyNmST)

Choose The Contribution Type You Are Using For Your Review: Algorithms/General
Justify The Above Choice:
I agree

Paper Summary:
This paper proposes SPAR, a unified MLLM framework designed to address the inherent conflict between semantic features and pixel features during unified visual understanding and generation. The authors introduce three core designs: (1) A dual-stream unified tokenizer, which uses an asymmetric design to decouple semantic retention (frozen InternViT) and pixel reconstruction (a newly added 6-layer Transformer + DC-AE decoder); (2) Dynamic Token Routing which uses a lightweight MLP to adaptively fuse multi-layer features based on token semantics to guide generation (3) A self-aligned generation paradigm, which directly uses the pre-trained dual-stream tokenizer as an internal alignment teacher for the DiT via feature distillation. Experiments demonstrate that SPAR achieves SOTA performance across multiple benchmarks, including image reconstruction, multimodal understanding, text-to-image generation, and image editing.

Paper Strengths:
- The most significant advantage of this work is its exceptionally strong experimental results. SPAR achieves leading scores on generation and editing benchmarks like GenEval (0.91), WISE (0.64), and ImgEdit (4.01), while reaching an impressive 0.27 rFID for ImageNet reconstruction. Notably, it gains robust generation capabilities without sacrificing original visual understanding performance (even showing slight improvements on benchmarks like MMB).
- The asymmetric dual-stream tokenizer is intuitive and engineering-effective solution that successfully mitigates catastrophic forgetting during joint training.
- The ablation studies in Tables 5 and 6 clearly demonstrate the necessity of the dual-stream design, the DTR mechanism, and the self-alignment loss, forming a complete logical loop.

Major Weaknesses:
- The DTR design is very similar to the commonly used MoE design. What's the unique advantage of this compared to MoE?
- The paper claims that its self-alignment mechanism "eliminates the dependency on external representation learners". however, the semantic stream of its tokenizer is fully anchored to the frozen, externally pre-trained InternViT model. The so-called "internal teacher" is still fundamentally built upon this externally pre-trained semantic encoder, meaning it does not truly eliminate external dependencies at all Furthermore, can this operation accelerate convergence speed in the same way that REPA does?

Minor Weaknesses:
- The training of the entire system is split into a staggering 6 stages (3 for the tokenizer + 3 for the unified model). This extremely complex stage-wise training not only incurs high hyperparameter tuning costs but also severely harms the reproducibility of the work.

Preliminary Recommendation: 4: Borderline Accept
Justification Of Preliminary Recommendation:

This paper is a typical system-level engineering work. Although at the fundamental algorithm level it leans towards a combination of existing mature techniques (feature decoupling, multi-layer routing, feature distillation) and slightly overclaims its novelty, it is undeniable that the authors have built an exceptionally powerful unified multimodal system. The dual-stream decoupling idea is proven to be highly effective in practice, providing the community with a strong and comprehensively leading baseline.

Suggestions For Rebuttal:
See weaknesses

Ethics Review Flag: No
Confidence Level: 4: High Confidence - The reviewer has strong expertise in the area. They are highly familiar with the relevant literature and can critically evaluate the paper.

Clear and Effective Design for Unified Multimodal Understanding and Generation

Official Review by Reviewer umgx 21 Apr 2026, 13:39 (modified: 03 May 2026, 06:52)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers, Authors
Revisions (/revisions?id=e8z37HjqjC)

Choose The Contribution Type You Are Using For Your Review: Algorithms/General
Justify The Above Choice:
I agree

Paper Summary:
The paper introduces SPAR to bridge the gap between semantic space and generative space without losing critical generative or semantic knowledge. First, they introduce a dual-stream asymmetric unified tokenizer, which utilizes both a semantic and pixel stream for capturing semantics and fine-grained image information. Next, they propose a framework to integrate the unified tokenizer for unified multimodal understanding and generation. Specifically, they present a Dynamic Token Routing mechanism to learn to leverage intermediate MLLM features for conditioning a downstream DiT. Furthermore, they propose using the unified tokenizer for aligning the intermediate representations of the DiT. Experiments on understanding and generation benchmarks show notable improvements.

Paper Strengths:
- The paper is simple and clear to follow. Each component is intuitive and fits well together, with the dual-stream design, dynamic token routing, and self-alignment paradigm forming a coherent pipeline.
- The DTR design is a compelling way for leveraging intermediate MLLM features as a conditioning for the downstream DiT.
- The empirical evaluation is comprehensive, covering reconstruction, understanding, generation, and editing.

Major Weaknesses:
- While overall performance is strong, it's hard to attribute where the performance gains are actually coming from. Some key ablations are highlighted below.
- Although the motivation behind DTR is compelling, there are limited ablations demonstrating if it actually leverages information from different layers.
- It's unclear what the impact of each stage of training is for the unified tokenizer. It would be beneficial to show ablations with just stage 1, then stage 1 + 2 and so on. If the semantic encoder permanently destroys high-frequency pixel information through lossy compression, it is hard to see how subsequent transformer layers can recover that information back, resulting in stage 3 being critical for downstream performance.
- Further ablations on the impact of unified tokenizer design on generation benchmarks such as GenEval and WISE would help better understand its impact on the model's generation capabilities.

Minor Weaknesses:
- It would be useful to compare training data size, number of extra parameters, and training time across different unified multimodal models, to better understand SPAR's efficiency relative to its performance gains.
- Although, [1] is already included in the paper, more direct discussion with [1] would help showcase the performance improvement with the proposed approach, as [1] also builds on top of InternVL3-2B.

References

Wu, Size, et al. "Openuni: A simple baseline for unified multimodal understanding and generation." *arXiv preprint arXiv:2505.23661* (2025).

Preliminary Recommendation: 5: Weak Accept
Justification Of Preliminary Recommendation:
Overall, the paper's motivations and core questions are compelling. The design is mostly clear and the performance gains are consistent across both understanding and generation benchmarks.

Suggestions For Rebuttal:
It would be appreciated if the authors could provide a clearer breakdown of how each training stage affects downstream performance. Furthermore, more ablations or visualizations regarding DTR would help clarify how it helps generation capabilities.

Ethics Review Flag: No
Confidence Level: 3: Moderate Confidence - The reviewer is reasonably knowledgeable about the topic. They understand the paper's methodology and results but may not be a leading expert in the specific subfield.

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